

TECHNOLOGY STACK

FitBuddy AI Fitness Plan Generator - Gemini Models

Date		22.09.2026	
Team ID		SWTID-2026-5226	
Project Name		Fitbuddy AI Fitness Plan Genetor Gemini Modls	
Maximum Marks		4 Marks	
Team Leader	Member	Member	Member
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Team Members

Define Your Technology Stack

The following technology stack is proposed for building the FitBuddy solution. Each choice supports the project requirements for an AI-powered, personalized fitness planning application.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React, HTML5, CSS3	Responsive and interactive interface. React supports reusable components; HTML/CSS structure and style fitness forms, plans, and progress views.
2	Backend / Server-Side	Python with FastAPI	Lightweight API layer for authentication, user inputs, plan requests, validation, and Gemini API communication.
3	Database / Data Storage	MongoDB	Flexible storage for user profiles, fitness preferences, generated plans, feedback, and progress records.
4	Cloud / Hosting / Deployment	Firebase Hosting + Render	Firebase Hosting serves the frontend; Render hosts the backend API for practical deployment of client and server components.
5	Version Control & CI/CD	GitHub + GitHub Actions	Supports team collaboration, version control, issue tracking, and automated testing/build workflows.
6	Third-Party APIs / Other Tools	Google Gemini API	Allows FitBuddy to send user fitness requirements to Gemini models and receive AI-generated personalized workout plans and fitness guidance.